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CSA0609- Design and Analysis of Algorithms

Champagne Tower

AIM

To write a Python program to find the amount of champagne in a particular glass of a champagne tower.

ALGORITHM

1. Start.
2. Read the poured champagne, row number, and glass number.
3. Initialize the first glass with the poured amount.
4. Distribute the excess champagne equally to the two glasses below.
5. Repeat until the required row is reached.
6. Display the amount of champagne in the required glass, limited to 1 cup.
7. Stop.

PSEUDOCODE

```
Begin
   Read poured, query_row, query_glass
   Initialize tower with zeros
   Set tower[0][0] = poured

   For each row:
      For each glass:
         If amount > 1:
            Distribute excess equally
   Print minimum of 1 and required glass amount
End
```

PYTHON CODE

main.py

Visualize

Run

```
p = float(input("Poured: "))
r = int(input("Row: "))
g = int(input("Glass: "))

a = [[0] * 101 for _ in range(101)]
a[0][0] = p

for i in range(r):
    for j in range(i + 1):
        if a[i][j] > 1:
            x = (a[i][j] - 1) / 2
            a[i+1][j] += x
            a[i+1][j+1] += x

print("Champagne:", min(1, a[r][g]))
```

OUTPUT

Output

```
Poured: 2
Row: 1
Glass: 0
Champagne: 0.5
```

COMPLEXITY ANALYSIS

Let r be the queried row.

Time Complexity: $O(r^2)$

Space Complexity: $O(r^2)$

RESULT

Thus, the Python program successfully calculates the amount of champagne present in the specified glass using a Dynamic Programming approach.